

Ishan Sinha

(224) 600-8497 | prof.ishan.sinha@gmail.com | [linkedin.com/in/sinhai/](https://www.linkedin.com/in/sinhai/) | github.com/ishansinha5 | ishansinha5.github.io/ishan-sinha-portfolio/

EDUCATION

Michigan State University

East Lansing, MI

Bachelor of Science in Computer Science | Concentration: Artificial Intelligence

Aug. 2024 – Dec. 2027

– Minors: Data Science; Computational Math, Science & Engineering; Mathematics | **Michigander Scholar**

EXPERIENCE

Auto-Owners Insurance

Lansing, MI

Software Development Intern

May 2026 – Aug. 2026

- Deploying configuration updates within the enterprise Guidewire Claim Center ecosystem, utilizing a proprietary object-oriented tech stack.
- Troubleshooting and optimizing backend software configurations for live production environments, ensuring high reliability for large-scale insurance claims processing.

Facility for Rare Isotope Beams (FRIB)

East Lansing, MI

Undergraduate Researcher

Jan. 2026 – Present

- Developed a 3D sparse neural network framework, reducing the computational cost of analyzing high-dimensional 3D structures.
- Designed the model to utilize sparse tensor operations, significantly reducing latency and compute overhead.

MSU Unmanned Systems

East Lansing, MI

Software Team Lead

Nov. 2025 – Present

- Directed the software engineering lifecycle for autonomous drone missions, from supervising flight path algorithms to training a proprietary machine learning model for aerial object detection.
- Programmed and validated autonomous flight paths by analyzing ArduPilot telemetry data to ensure mission compliance.

Encora

Remote

Software Development Intern

Jul. 2023 – Aug. 2023

- Created a polling web application utilizing full stack application approaches, through the Django Python framework.
- Built a data processing module to rank and filter user-generated poll data in real-time.

PROJECTS

NBA Finals NLP Predictor | *Python, Hugging Face (RoBERTa), Streamlit*

[GitHub](#)

- Engineered an automated data ingestion pipeline processing post-game transcripts (2018-2025) to isolate and analyze temporal sentiment shifts across coaches, star players, and role players.
- Launched a live predictive dashboard tracking team momentum and opponent reaction metrics to forecast playoff outcomes.
- Automated the video ingestion and enhancement process to broaden dataset coverage, significantly increasing model predictive accuracy for championship-level mentalities.

Comic Book Shop Detector & Recommender | *Python, Java (Spring Boot), C++, PostgreSQL, Vercel*

[GitHub](#)

- Launched a full-stack spatial locator application, integrating a Java Spring Boot and PostGIS backend to process user location data against the Google Places API.
- Architected a native C++ Perceptual Hashing (pHash) microservice to bypass expensive GPU use by retrieving cached comic data, falling back to a PyTorch/FastAPI vision inference layer for novel images.
- Constructed a comic release schedule service by web scraping publisher data and integrating the Google Calendar API via a Flask backend, automating personal calendar updates.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: PyTorch, Scikit-Learn, Hugging Face, Spring Boot, Flask, Node.js, Pandas, NumPy

Technical Proficiencies: Unsupervised Learning, Autonomous Systems, Real-time Object Detection, Telemetry Analysis

Developer Tools: Git, GitHub, GitHub Actions, Docker, CI/CD Pipelines, REST APIs, Linux/Unix